

Q. Does the scatterplot show strong evidence of a linear relationship between the data? Based on your answer, would you recommend a linear classifier or something more complex to best fit this data?

A. After carefully examining the scatter plot matrix and considering the patterns in the data. No, the scatter plot doesn't show strong evidence of a linear relationship between the data. So, I would recommend using a more complex model rather than a linear classifier to best fit this data

Q. Write a brief (1-2 sentences) summary describing each feature to demonstrate your knowledge of key features used in this model. (4 points)

- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape
- Marginal Adhesion
- Single Epithelial Cell Size
- Bare Nuclei
- Bland Chromatin
- Normal Nucleoli
- Mitoses
- Class

A.

Clump Thickness : measures the uniformity of cell clumps, with higher values indicating more irregular clumping, which can be indicative of malignancy.

Uniformity of Cell Size : It quantifies the variation in size among cells greater uniformity suggests benign characteristics while variability may point to malignancy.

Uniformity of Cell Shape :assesses the consistency in shape across cells

Marginal Adhesion : It evaluates how well cells stick to each other at the edges, increased adhesion is associated with cancerous growths.

Single Epithelial Cell Size : measures the size of individual epithelial cells, larger sizes are often linked to malignant conditions.

Bare Nuclei : indicates the presence of nuclei that are not surrounded by cytoplasm, a characteristic more common in malignant cells.

Bland Chromatin : This feature describes the texture of chromatin within the nucleus; blandness typically signifies benign tumors.

Normal Nucleoli : It assesses the normalcy of nucleoli (structures within the nucleus); abnormal nucleoli are often found in cancerous cells.

Mitoses : This counts the number of dividing cells; a higher mitotic rate is usually indicative of rapid cell division seen in malignant tumors.

Class : This is the target variable that categorizes samples as either benign (2) or malignant (4).

Q. Explain why it is important to evaluate your model over the validation dataset and not only test dataset?

A. The purpose of the validation dataset is to fine tune the model during the training process. It helps in selecting the best model architecture, hyperparameters. By evaluating the model on the validation dataset during training, you can monitor the model's performance and detect signs of overfitting early. The validation dataset provides an independent evaluation of the model's performance during the training phase. This helps in making informed decisions about when to stop training, which model to select, and how to improve the model.

Then we can use the test set as a final, independent evaluation of the model's performance on unseen data. This leads to more reliable and robust models that perform well in real-world scenarios.

Q. Write a brief (1-2 sentences) summary of each measure and comment about each measure's score in your assessment of the two models

Summary of Each Measure

- Accuracy : It measures the proportion of correctly classified instances out of all instances.
- Precision : It's the ratio of true positive predictions to the total predicted positives. It indicates how many of the predicted positive instances are actually positive.
- Recall : Recall, also known as sensitivity, is the ratio of true positive predictions to the total actual positives. It shows how many of the actual positive instances were correctly identified.
- F1-score : The F1-score is the harmonic mean of precision and recall, providing a balanced measure of both. It is particularly useful when there is an uneven class distribution.

Comment on Each Measure's Score

- KNN Model :
 - Validation Set : The KNN model shows strong performance with high scores across all metrics (accuracy: 0.9571, precision: 0.9570, recall: 0.9571, F1-score: 0.9569). This suggests that the model is effectively capturing the patterns in the data during validation.
 - Test Set : The KNN model performs exceptionally well on the test set, achieving near-perfect scores (accuracy: 0.9857, precision: 0.9857, recall: 0.9857, F1-score: 0.9857). This indicates excellent generalization and robustness of the model on unseen data.
- SVM Model :
 - Validation Set : The SVM model shows very good performance with consistent scores (accuracy: 0.9714, precision: 0.9714, recall: 0.9714, F1-score: 0.9714). This indicates that the model is effectively learning from the training data and validating well.
 - Test Set : The SVM model maintains its strong performance on the test set, with slightly varied but still high scores (accuracy: 0.9714, precision: 0.9715, recall: 0.9714, F1-score: 0.9713). This suggests that the model generalizes well to new data, though it may not be as robust as the KNN model based on the test set results.

In summary, both models exhibit strong performance across all evaluation metrics, with the KNN model showing a slight edge in terms of generalization on the test set.